MILITARY SPECIFICATION

TRAILER, AIRCRAFT ENGINE (ETU-91/E)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an aircraft engine trailer, designated ETU-91/E.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements For

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-281	Automobiles, Trucks, Truck-Tractors, and Trailer Dollies, Preservation and Packaging Of
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-882	System Safety Program Requirements
* MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center/MMEDO, Kelly AFB, Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1740

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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

United Aircraft Products, Incorporated

UA533560-2 Trailer, Aircraft Engine ETU-91/E

* 2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951-82 Packaging, Commercial

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 General requirements. The requirements of United Aircraft Products, Inc. drawing UA533560-2 apply as requirements of this specification.

3.3 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.3.1 Safety. To the extent specified by the procuring activity, the provisions of MIL-STD-882 shall apply.

3.4 Workmanship. The trailer, including all parts and accessories shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, sharp edges, accuracy of dimensions, and markings of parts and assemblies.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the trailer shall be classified as follows:

a. First Article Inspection. (See 4.4)

b. Quality Conformance Inspection. (See 4.5)

4.3 Test conditions.

4.3.1 Standard test conditions. Unless otherwise specified herein, all tests shall be conducted under room conditions with the temperature at 25° ±10°C, the relative humidity at 90 percent or less, and the barometric pressure at 28 to 32 inches Hg.

4.3.2 Apparatus and instruments. Apparatus and instruments used in conjunction with the testing specified herein shall be of laboratory precision type, checked to ascertain that they have been installed for durability and accurate operation, and calibrated at intervals properly spaced to assure laboratory efficiency.

4.3.3 Mockup engine. A dead weight load equivalent to the configurational dimensions, 12,000 pound capacity, center-of-gravity position, and mounting provisions of the bare engine load shall be provided and utilized for test purposes. The bare engine load shall be handled by sling and attachment fittings simulating those of the actual engine.

4.3.4 Data. During the testing specified herein, sufficient data shall be recorded at proper intervals to insure compliance of the trailer with the performance requirements specified. Recorded data shall include the date, time, and ambient temperature at which the test is conducted as well as the data needed to accurately calculate the performance rates, efficiencies, et cetera, to be determined by the specific tests.

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4.3.5 Failure during test. Characteristics exhibited by the trailer during any of the tests specified herein that would cause the trailer to fail any of the requirements specified herein while in service shall constitute failure of the unit to pass the test and shall be the basis for rejection.

4.4 First Article Inspection.

4.4.1 Test sample. One trailer shall be subjected to the first article tests specified in 4.4.4.

* 4.4.2 Test report. Upon completion of the first article tests, a first article test report in accordance with MIL-STD-831 shall be prepared and three copies of the report furnished to the procuring activity.

* 4.4.2.1 Reliability and maintainability information. The following information shall be submitted as a separate attachment accompanying the first article test report (same quantity and quality of copies) or shall be included as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulating operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during failures and irregular operations shall be recorded.

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failures and irregular operations.

c. A summary of the engineering analysis behind any corrections made, to be made on production units, or proposed to be made and behind the predicted effectiveness of those corrections.

d. Clock time and manhours required for each maintenance and servicing action during the tests. A brief description of the qualification and experience of the personnel involved shall be included and shall be adequate to allow comparison to the personnel anticipated in similar field work.

* 4.4.3 First Article test sample for the procuring activity. After submission of the first article test report, the contractor shall retain the preproduction test sample at his plant in operating condition pending the following actions by the procuring activity:

a. Review of the mechanical construction

h. Performance of any test, included in the specification, after reviewing the contractor's test report.

c. Notification of final acceptance.

* 4.4.4 First Article tests. The first article tests shall consist of all the tests described under 4.6, conducted in the same order.

4.5 Quality conformance tests. The quality conformance tests shall consist of the following:

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a. Individual tests (See 4.5.1)

b. Sampling tests (See 4.5.2)

4.5.1 Individual tests. Each trailer shall be subjected to the following tests as described under 4.6:

a. Examination of product (See 4.6.1)

b. Operational tests (See 4.6.2)

4.5.2 Sampling test. One trailer shall be selected at random from every ten or fraction thereof and subjected to 10 percent of the test specified in 4.6.6.

4.5.2.1 Rejection and retest. When one trailer selected from a production run fails to meet the specification, trailers still on hand or produced later shall not be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative and notify the procuring activity in writing the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.1.1 Individual tests may continue. For operational reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the trailers on hand or produced later shall not be made until it is determined that all trailers meet all the requirements of the specification.

4.5.2.1.2 Defects in trailers already accepted. The investigation of a test failure could indicate that defects may exist in trailers already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The trailer shall be examined to determine conformance to the requirements of this specification with respect to dimensions, quality of workmanship, materials, design of finishes, markings, and visible defects, and the functional operation of all components, assemblies, accessories, and controls. Particular attention shall be given to leakage in the hydraulic system, and the routing and protection of hydraulic lines.

4.6.2 Operational tests. The trailer shall be operated under the following conditions to insure that the various components and capabilities comply with the requirements specified herein. During this performance demonstration, all components shall be operated according to the instructions provided with the trailer.

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4. 6.2.1 Engine installation and removal. The trailer length shall be adjusted to the retracted position and the mockup bare engine configuration shall be hoisted into position on the trailer. The engine shall be mounted on the trailer with particular attention given to the compatibility of the trailer mounting supports and engine fixtures. The lift mechanism shall be operated through the entire range of travel and the safety lock capability shall be demonstrated.

4.6.3 Mobility tests. While supporting the rated capacity load, the trailer shall be tested for type I, group C mobility in accordance with MIL-M-8090, with the following exceptions:

4.6.3.1 Towing distances. The towing distances shall be 50 miles over level paved highway type surface and 50 miles over the type I test course. Fifty percent of each test distance shall be with the trailer in the extended position and the remaining fifty percent in the retracted position.

4.6.3.2 Towing speeds. Towing speeds for the paved roads shall be 10 $\pm$1 mph and 5 $\pm$1 mph for the type I test course.

4.6.3.3 Rear steering. The manual operation and lockout capabilities of the rear steering mechanism shall be demonstrated.

4.6.4 Excessive load indicator test. A test shall be performed to demonstrate that the excessive load indicator provides a visual indication that loads of 3.5g or greater have been transmitted to the mounted engine during mobile operations.

4.6.5 Static proof load tests.

4.6.5.1 A static proof down load of 24,000 pounds shall be applied to the trailer through the engine mounting supports. The trailer length shall be extended; the lift mechanism shall be centrally positioned; and the safety locks shall be disengaged. Footpads shall be clear of the ground line. The load shall be maintained for 3 minutes without evidence of leakage, instability, or permanent deformation of any component or part during or after completion of the test.

4.6.6 Endurance tests. The lift mechanism shall be operated through its entire range capability for 50 cycles, while the trailer is supporting the bare mockup engine. At the completion of every fifth cycle, the engine shall be removed and reinstalled on the trailer mounting supports. The trailer extension and retraction mechanism shall then be operated through 50 cycles. At the completion of every fifth cycle, and while the trailer is in the retracted position, the bare mockup engine shall be installed and removed from the trailer mounting supports. Failure, deterioration, undue wear, or unsatisfactory operation resulting from the endurance tests shall be cause for rejection.

4.6.6.1 Leakage inspection. During the endurance tests, the trailer shall be inspected for leakage. Fluid lines, fittings, pumps, and other components shall show no leakage. However, if pumps and other equipment have packed glands that can be tightened during the endurance test to stop leakage that may have occurred, such adjustments will be acceptable.

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4.6.7 Reliability demonstration and test. Satisfactory completion of all tests specified herein shall be a demonstration of compliance with the quantitative reliability requirements of this specification.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PACKAGING.

* 5.1 Preservation. Presentation shall be level A, C or commercial, as specified (see 6.2).

* 5.1.1 Level A. The trailer shall be preserved and packaged in accordance with MIL-STD-281.

* 5.1.2 Level C. The trailer shall be preserved and packaged in accordance with MIL-STD-281.

* 5.1.3 Commercial. The trailer shall be preserved in accordance with ASTM D 3951.

* 5.2 Packing.

* 5.2.1 Packing. Packing shall be level A, B, C, or commercial, as specified (see 6.2).

* 5.2.2 Level A. The trailer shall be shipped unboxed mobil as specified in MIL-STD-281 for levels A.

* 5.2.3 Level B. The trailer shall be shipped unboxed mobil as specified in MIL-STD-281 for level B.

* 5.2.4 Level C. The trailer shall be shipped unboxed mobil as specified in MIL-STD-281 for level C.

* 5.2.5 Commercial. The trailer shall be shipped unboxed mobil in accordance with requirement of ASTM D 3951.

* 5.3 Physical Protection. All cushioning, blocking, bracing, and bolting shall be in accordance with MIL-STD-1186. The drop test of MIL-STD-1186 shall be waived when preservation, packaging and packing of the trailer is for immediate use or when the drop tests of MIL-P-116 are applicable.

* 5.4 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129, or ASTM D 3951 as applicable.

6. NOTES

6.1 Intended use. The ETU-91/E trailer is intended for transportation, storage, installation, and removal operations for the Type C-5A aircraft TF-39 turbofan engine.

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6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. That MIL-M-8090E(USAF), dated 18 April 1966, or later issue be used in conjunction with this specification.

c. Location and conditions for first article (see 4.4).

d. Levels of preservation, packaging, and packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of' the documents first article.

6.4 Definitions. For the purpose of this specification, the following definitions will apply:

6.4.1 Bare engine. The bare engine configuration of the TF-39 engine is defined as the core engine (compressor stage, combustion and tail pipe sections) with the fan section installed.

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-83229B (USAF)	Trailer, Aircraft Engine ETU-91/E

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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